

Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Project Title

MediLink – Smart Doctor Appointment & Healthcare Management System.

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MediLink – Smart Doctor Appointment & Healthcare Management System.



Figure : Hospital Management System (MediLink) App Diagram

Description:

The use case diagram of the Medilink system illustrates the interaction between the main actors: Patient, Doctor, and Admin with the system functionalities. The system is designed to provide a smooth and efficient healthcare management platform.

Patients can register into the system, log in securely, search for doctors based on specialization, and book appointments according to available time slots. Doctors can log in to their dashboard, manage their schedules, and accept or reject appointment requests from patients. Admin has full control over the system, including managing users, doctors, and monitoring system activities.

This diagram clearly represents the overall functionality of the system and shows how each user interacts with different features of the Medilink platform.

Login Screen



Figure : Login Screen

Register Screen

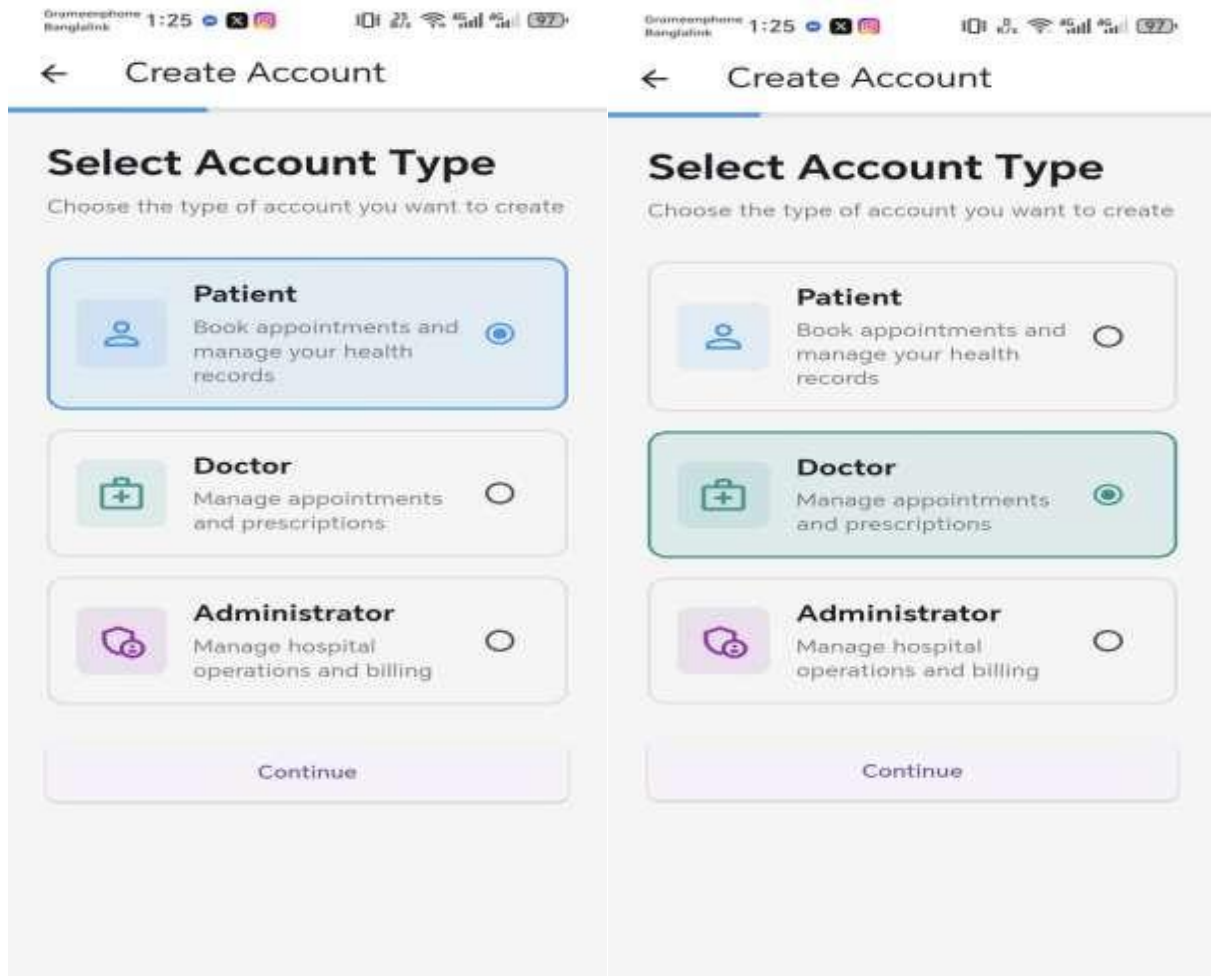


Figure : Register Screen

The register screen enables new users to create an account in the MediLink App by providing essential details such as name, email address, contact information, and password. The registration process is integrated with Firebase Authentication to ensure secure and reliable user account creation. The screen includes basic input validation to check for missing or incorrect information and provides feedback messages to guide users during the registration process. After successful registration, user data is stored in the database and the user is assigned a role, which determines their access level within the system. This screen plays a key role in maintaining data accuracy, system security, and smooth onboarding of new users.

Patient Dashboard

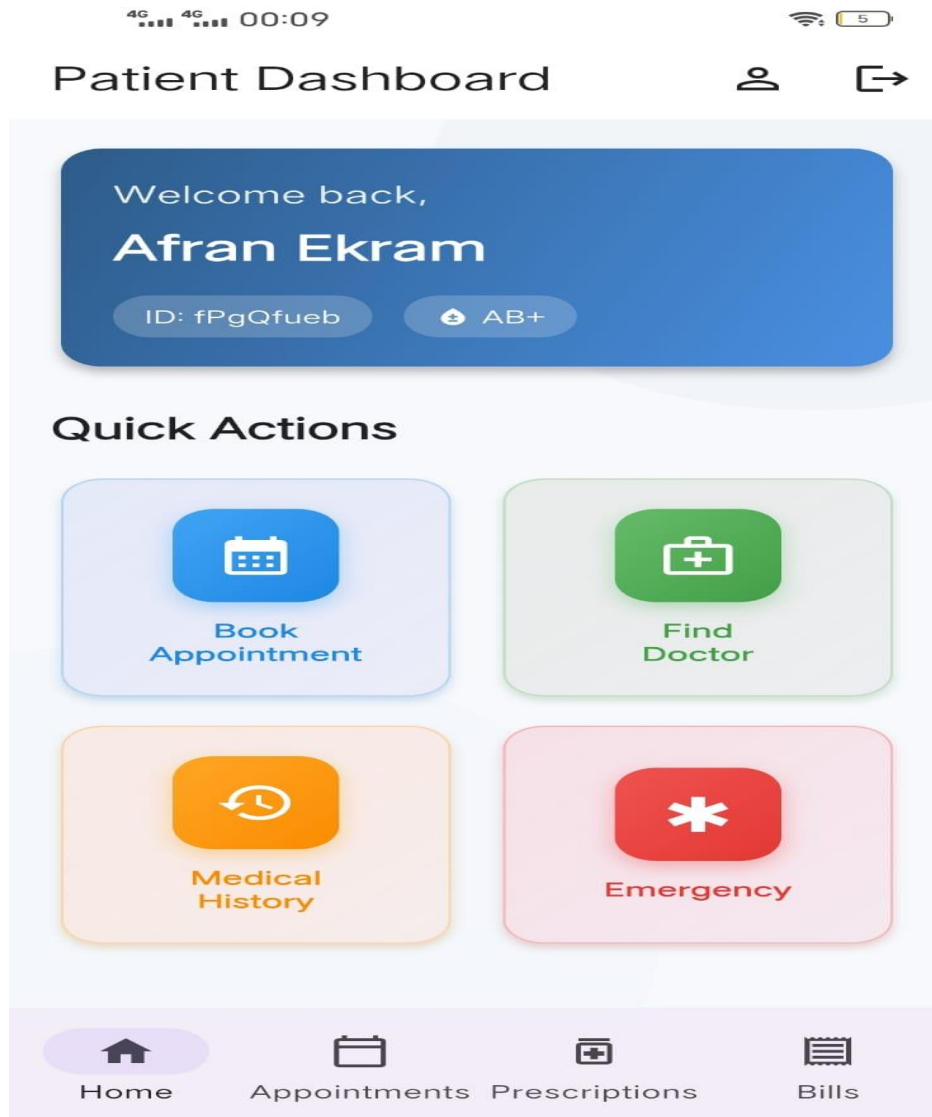


Figure : Patient Dashboard

The patient dashboard serves as the main interface for patients after successful login into the MediLink App. It provides a centralized view of key features such as appointment booking, viewing upcoming and past appointments, accessing medical records, and checking billing information. The dashboard is designed with a clean and user-friendly layout to ensure easy navigation and quick access to important services. Real-time updates from the Firebase backend allow patients to receive instant notifications about appointment confirmations, schedule changes, and system alerts. This screen enhances patient engagement and improves communication between patients and healthcare providers.

Doctor Dashboard

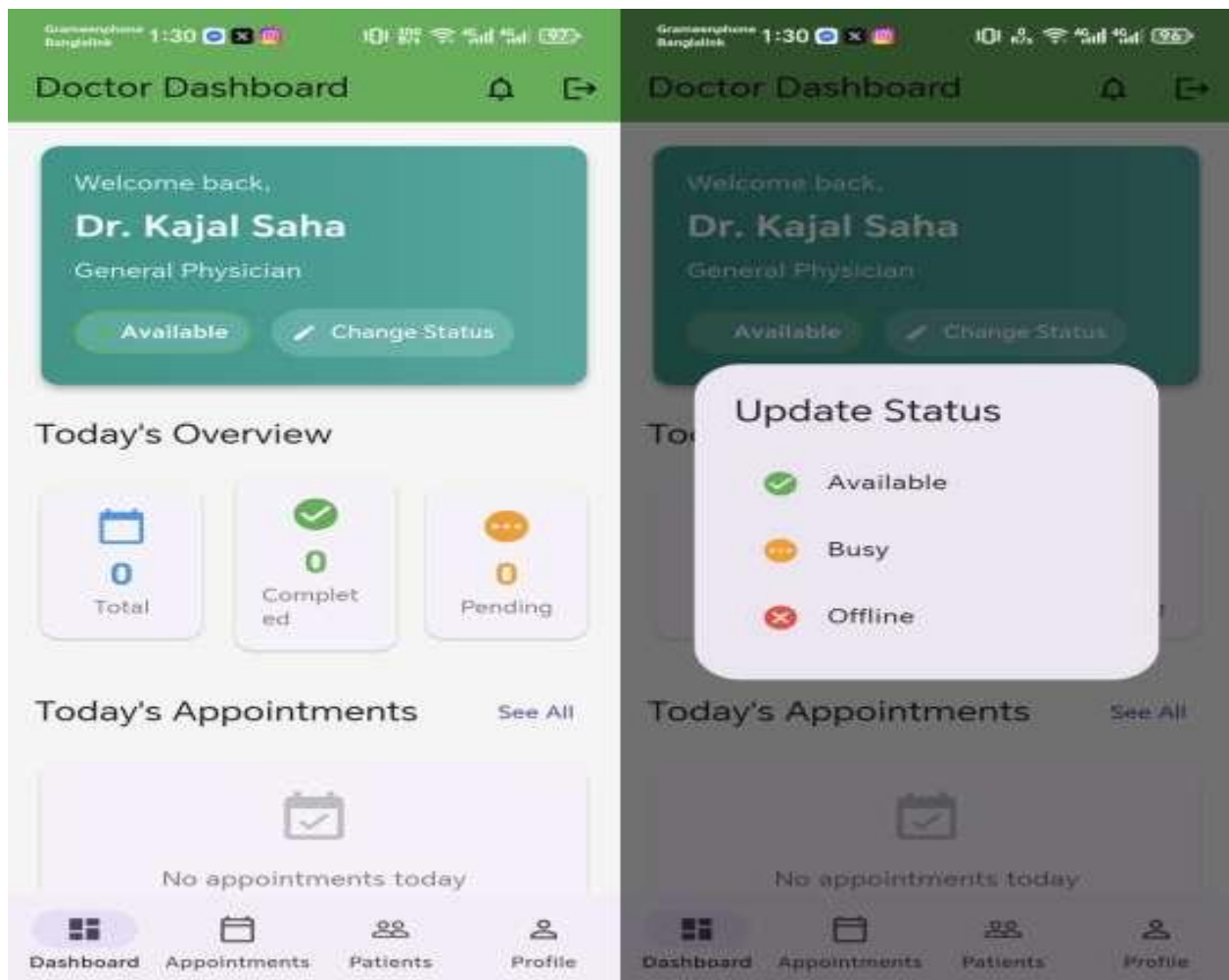


Figure : Doctor Dashboard

The doctor dashboard provides a centralized interface for doctors to manage their daily activities within the MediLink App. It displays scheduled appointments, patient details, and access to medical records in an organized manner. Doctors can view patient histories, update consultation notes, and manage their availability through this screen. Realtime data synchronization via Firebase ensures that any updates made by the doctor are instantly reflected across the system, improving coordination and enhancing the quality of patient care.

Book Appointment

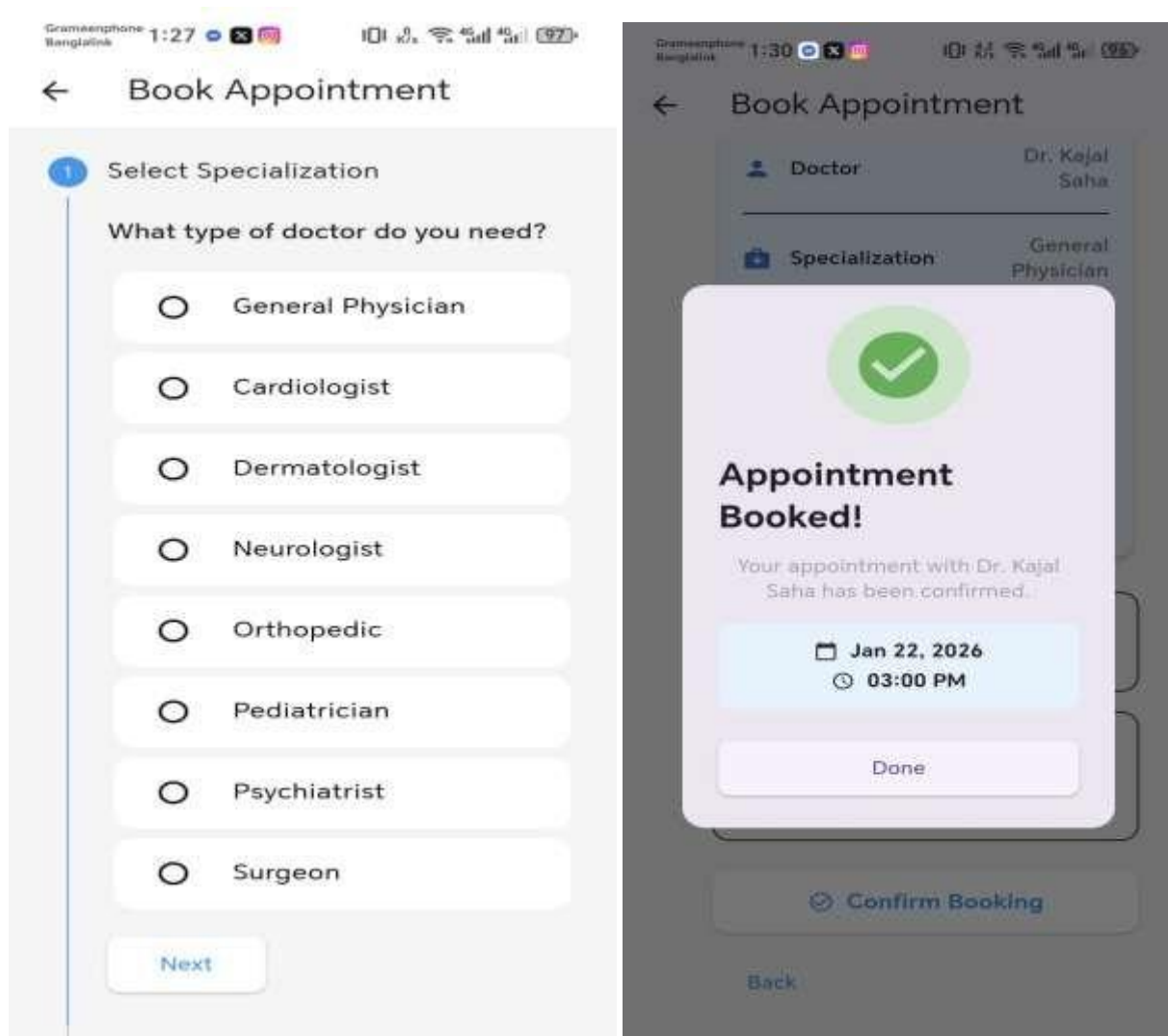


Figure : Book Appointment

The Book Appointment screen allows patients to schedule consultations with doctors through a simple and guided process. Patients can select the desired department, choose an available doctor, and pick a suitable date and time slot based on real-time availability. The system validates the input data and stores the appointment details securely in the Firebase database. Once the booking is successful, confirmation messages and notifications are generated for both the patient and the doctor. This feature reduces manual scheduling efforts, minimizes waiting time, and improves overall coordination within the hospital system.

Logout

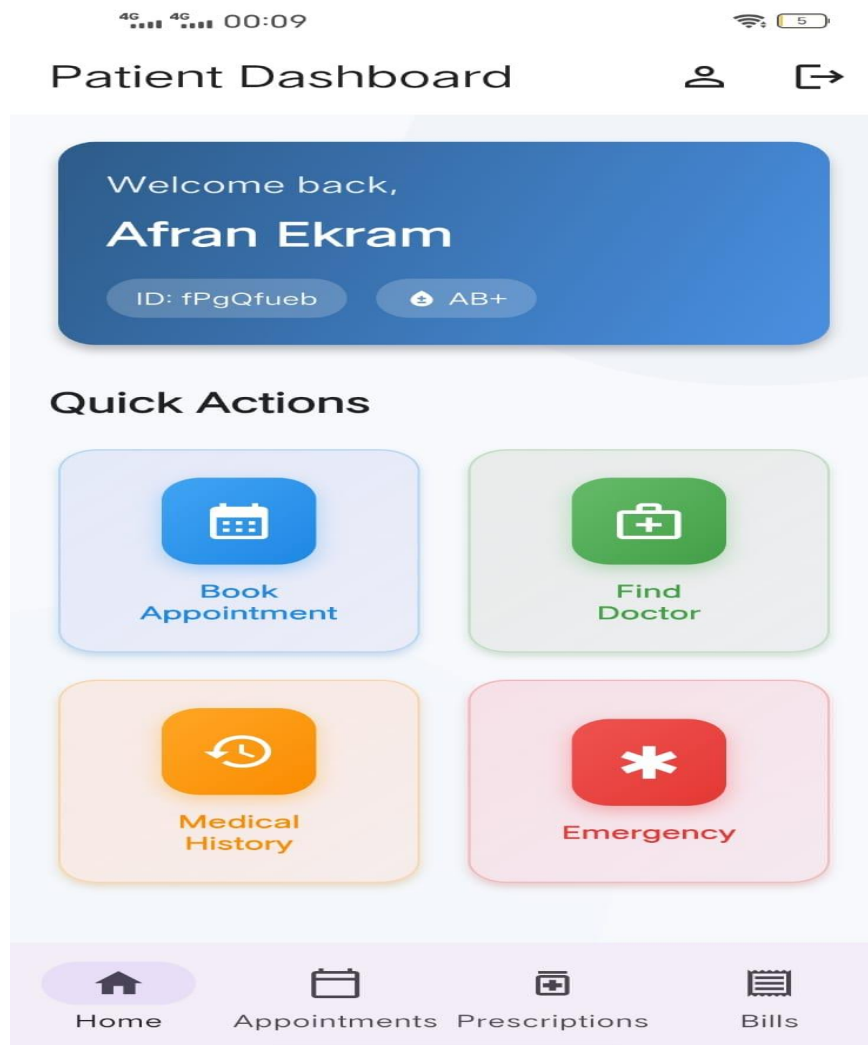


Figure : Logout

The Logout feature allows users patients, doctors, and administrators to securely exit the MediLink App after completing their tasks. By signing out, the user's session is terminated, preventing unauthorized access to sensitive information. This function is integrated with Firebase Authentication to ensure that all user sessions are properly managed. The logout process helps maintain data security, protects privacy, and ensures that the application remains safe for multi-user access on shared devices.

➤ Landing/Home Page

Features:

- System Introduction
- Login Button
- Register Button
- Services Overview

- Contact Information

➤ Login Page

Components:

- Email Field
- Password Field
- Forgot Password
- Login Button
- Register Redirect

➤ Registration Page

Components:

- Full Name
- Email
- Phone Number
- Password
- User Type (Patient/Doctor)
- Register Button

➤ Patient Dashboard

Features:

- Book Appointment
- Medical Records
- Prescriptions
- Notifications
- Profile Management

➤ Doctor Dashboard

Features:

- Appointment Schedule
- Patient History
- Prescription Management
- Video Consultation
- Availability Settings

➤ Admin Dashboard

Features:

- User Management
- Doctor Management

- Appointment Monitoring
- Payment Monitoring
- Analytics Dashboard

➤ Appointment Page

Features:

- Doctor Search
- Date Selection
- Time Slot Selection
- Confirm Appointment

➤ AI Symptom Checker Page

Features:

- Symptom Input
- Disease Prediction
- Doctor Recommendation
- Health Suggestions

➤ Profile Page

Features:

- Edit Profile
- Change Password
- View Medical History
- Upload Reports

➤ Design Guidelines:

- Responsive Design
- Consistent Color Scheme
- Easy Navigation
- User-Friendly Interface
- Mobile First Design

➤ Technology Used for Design:

- Figma
- Draw.io